



Complete Vegetable Storage & Intelligence Report

Reporting Period: 24h

OPERATOR: Dr. Aris Thorne (demo@vegsense.io)

DEVICE: ESP32-DEMO-001 [DEMO DATA]

TARGET BATCH: BATCH-2026-TOM-01 (12 kg)

TELEMETRY: 24 points (raw)

Executive Environmental Summary

AVG TEMPERATURE

27.6°C

Range: 26.2 - 29°C

AVG HUMIDITY

73.3%

Range: 70 - 76%

AVG GAS/VOC INDEX

419

Relative MQ-135 index

AVG LIGHT LEVEL

178 lx

LOW LIGHT

ESTIMATED RISK

17%

Status: FRESH

TOTAL ALERTS

9

0 critical, 2 active

MONITORED MASS

82 kg

0 active batches

DATA QUALITY

85%

FAIR

Atmospheric Sensor Monitoring Analysis

Historical statistics recorded across storage chamber sensor arrays.

SENSOR METRIC	CURRENT	AVERAGE	MINIMUM	MAXIMUM	CONDITION STATUS
Temperature (°C)	26.6°C	27.6°C	26.2°C	29°C	NORMAL
Relative Humidity (%)	73%	73.3%	70%	76%	OPTIMAL
Gas/VOC Indicator	412	419	411	431	NORMAL
Light Level (lux)	0 lx	178 lx	0 lx	450 lx	LOW LIGHT

Estimated Spoilage Risk & Factor Breakdown

Multivariate risk scoring derived from Part 6 preservation heuristics.

RISK INDEX SUMMARY:

Current: 16% | Average: 17% | Classification: FRESH | Historical Trend: STABLE

FACTOR CONTRIBUTIONS:

Thermal Impact: 15% | Moisture Impact: 14% | VOC Impact: 18% | Photometric: 10%

RISK SPECTRUM TIME:

Fresh (0–30%): 100% | Warning (31–60%): 0% | Risk (61–80%): 0% | Critical: 0%



Alert Notifications & Atmospheric Incidents

Total recorded: 9 (Critical: 0, High: 5, Warning: 2) | Avg Resolution: 9 min

Timestamp	Incident Title	Category	Severity	Status	Device
01:16 pm	Gas/VOC Sensor Data Unavailable	SENSOR_DATA_UNAVAILABLE	WARNING	RESOLVED	ESP32-DEMO-001
01:16 pm	Light Sensor Data Unavailable	SENSOR_DATA_UNAVAILABLE	WARNING	RESOLVED	ESP32-DEMO-001
01:16 pm	High Humidity	HUMIDITY	HIGH	RESOLVED	ESP32-DEMO-001
01:17 pm	Gas/VOC Sensor Recovered	SENSOR_RECOVERY	INFO	ACTIVE	ESP32-DEMO-001
01:17 pm	Light Sensor Recovered	SENSOR_RECOVERY	INFO	ACTIVE	ESP32-DEMO-001
01:17 pm	High Humidity	HUMIDITY	HIGH	RESOLVED	ESP32-DEMO-001
01:18 pm	High Humidity	HUMIDITY	HIGH	RESOLVED	ESP32-DEMO-001
01:29 pm	High Humidity	HUMIDITY	HIGH	RESOLVED	ESP32-DEMO-001

Vegetable Storage Inventory Status

Total inventory: 4 batches (82 kg) across 4 produce types.

Produce Variety	Batches	Estimated Risk	Alerts	Storage Condition
Tomato	1 batch(es)	18%	0	OPTIMAL
Potato	1 batch(es)	12%	0	OPTIMAL
Onion	1 batch(es)	14%	0	OPTIMAL
Carrot	1 batch(es)	19%	0	OPTIMAL

Actionable Storage & Engineering Recommendations

Targeted adjustments derived from active telemetry and threshold events.

ACTIVE ALERTS

Acknowledge and inspect storage bays corresponding to active alert triggers.

2 unresolved incident notifications currently flagged.

TECHNICAL DISCLAIMER & SYSTEM SCOPE NOTE

VegSense provides environmental monitoring and estimated storage-condition/spoilage risk based on available sensor and storage data. The Gas/VOC sensor is an environmental indicator and does not by itself identify a specific spoilage gas. Risk classifications are configurable estimates and should not be interpreted as definitive confirmation of physical spoilage.

